

Current And Resistance

CHAPTER - 26, P.H.N 10th Edn

⑥ Electric Current

last lectures we discussed electrostatics - The Physics of stationary charges.

now, we discuss the physics of electric currents - it represents change in motion.

~~II~~ ~~III~~ Here we discuss the basic physics of electric currents and why they can be established in some materials but not in others.

→ Currently we will discuss - currents of conduction electrons moving through metallic conductors such as copper wires.

Definition: If charge dq passes through a hypothetical plane/surface in time interval dt .

$$i = \frac{dq}{dt}$$

total charge for that time interval

$$q = \int dq = \int_0^t i dt \quad [\text{Note: 'i' may vary with time}]$$

The SI unit for current is the coulomb per second, or the ampere (A), which is SI base unit.

$$1 \text{ ampere} = 1 A = 1 \text{ coulomb per second} = 1 C/s$$

Charge is conserved

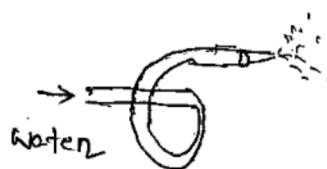
$$i_0 = i_1 + i_2$$



Directions of Currents

A current arrow is drawn in the direction in which positive charge carriers would move, ~~even~~ even if the actual charge carriers are negative and move in the opposite direction.

Sample problem 26.01



Water flows through a garden hose at a volume flow rate dv/dt of $450 \text{ cm}^3/\text{s}$. What is the current of negative charge?

$$i = (\text{per electron charge}) (\text{electrons per molecule}) (\text{molecules/second})$$

$$i = (e) (10) \left(\frac{dN}{dt} \right) \quad \text{--- (1)}$$

$$\frac{dN}{dt} = (\text{molecules per mole}) (\text{moles per unit mass}) \times (\text{mass per unit volume}) (\text{volume per second})$$

$$= N_A (\text{Avogadro's number}) \left(\frac{1}{M (\text{molar mass})} \right) \rho_{\text{mass}} \left(\frac{dv (\text{volume})}{dt} \right)$$

$$i = 10e N_A M^{-1} \rho_{\text{mass}} \frac{dv}{dt}$$

$$i = 24.1 \text{ MA}$$

$$\begin{aligned} N_A &\rightarrow 6.02 \times 10^{23} \text{ molecules} \\ \text{Water } \rho_{\text{mass}} &= 1000 \text{ kg/m}^3 \\ M &= 18 \text{ g/mol} \\ \frac{dv}{dt} &= 450 \times 10^{-6} \text{ m}^3/\text{s} \end{aligned}$$

Current Density

Current i (a scalar quantity) is related to current density $\vec{J}$ (a vector quantity) by $i = \int \vec{J} \cdot d\vec{A}$,

where $d\vec{A}$ is a vector perpendicular to surface element of area dA and the integral is taken over any surface cutting across the conductor.

[1] $\vec{J}$ has the same direction as the velocity of the moving charges if they are positive and the opposite direction if they are negative.

[2] When an electric field $\vec{E}$ is established in a conductor, the charge carriers acquire a drift speed $\vec{v}_d$ in the direction of $\vec{E}$.

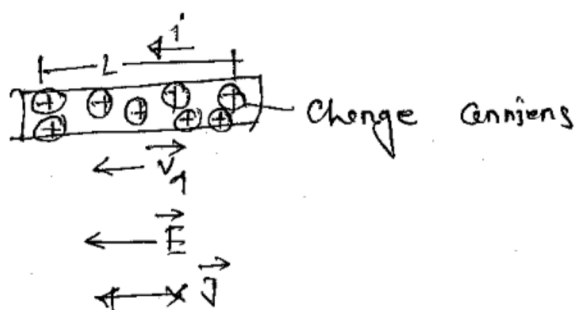
$\vec{v}_d$ is related as $\vec{J} = (ne) \vec{v}_d$.
 $\downarrow$
 Charge density.

Current Density: $i = \int \vec{J} \cdot d\vec{A}$

$J = \frac{i}{A} \text{ (A/m}^2\text{)}$ Amphere per square meter.

Drift Speed

(4)



Current is said to be due to positive charges that are propelled by the electric field.

Charge carriers move with the same drift speed v_d and J is uniform across the wire's cross-sectional area A .

$$Q = (nAL)e \quad \left| \begin{array}{l} n \rightarrow \text{number of carriers per} \\ \text{unit volume} \end{array} \right.$$

$$t = \frac{L}{v_d}$$

$$i = \frac{Q}{t} = \frac{nALe}{L/v_d} = nAe v_d$$

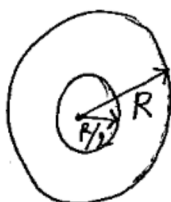
$$v_d = \frac{i}{nAe} = \frac{J}{ne}$$

$$\boxed{\vec{J} = (ne) \vec{v}_d}$$

Note $\vec{J}$ and $\vec{v}_d$ has the same direction

S-P 26.02

cylindrical wire



$R = 2.0 \text{ mm}$ (uniform)

$J = 2.0 \times 10^5 \text{ A/m}^2$

Q → What is the current through the outer portion of the wire between radial distances $R/2$ and R .

$$i = JA \quad \text{--- (1)}$$

$$A = \pi R^2 - \pi \left(\frac{R}{2}\right)^2 = \pi \left(\frac{3R^2}{4}\right) = 9.424 \times 10^{-6} \text{ m}^2$$

$$i = JA' = (2.0 \times 10^5 \text{ A/m}^2) (9.424 \times 10^{-6} \text{ m}^2) = 1.9 \text{ A}$$

(b) Suppose, instead, that the current density through a cross section varies with radial distance r as $J = ar^2$, in which $a = 3.0 \times 10^{11} \text{ A/m}^4$ and r in meters.

$$i = \int \vec{J} \cdot d\vec{A} \quad [\text{As current density is not uniform}]$$

$\vec{J}$ and $d\vec{A}$ have the same direction

$$\vec{J} \cdot d\vec{A} = \cancel{\vec{J}} \cdot \cancel{d\vec{A}} = J dA \cos 0 = J dA$$

$$i = \int \vec{J} \cdot d\vec{A} = \int J dA$$

$$= \int_{R/2}^R \frac{ar^2}{2} 2\pi r dr = 2\pi a \int_{R/2}^R r^3 dr$$

$$= 2\pi a \left[\frac{r^4}{4} \right]_{R/2}^R = \frac{\pi a}{2} \left[R^4 - \frac{R^4}{16} \right] = \frac{15}{32} \pi a R^4$$

$$= 7.1 \text{ A}$$

Resistance and Resistivity

Observation: If we apply the same potential difference between the ends of geometrically similar rods of copper and of glass, very different current result.

The characteristic of the conductor that enters here is its electrical resistance.

The resistance $R = \frac{V}{i}$ (definition of R)

$$\Rightarrow \frac{\text{Volt}}{\text{ampere}} = \text{ohm}.$$

A conductor whose function in a circuit is to provide a specified resistance is called a resistor. Symbol $\square$.

$i = \frac{V}{R}$, represents, for a given V , the greater the resistance, the smaller the current.

Resistivity: Now we count Electric field $\vec{E}$ associate with material resistance.

$$J = \frac{E}{\rho} \quad (\text{definition of } \rho) \quad \text{Unit} = \frac{V/m}{A/m^2} = \frac{V}{A} m$$

$$\boxed{\vec{E} = \rho \vec{J}} \quad [\text{isotropic materials}]$$

$$= \Omega \cdot m.$$

Conductivity σ of a material, $\sigma = \frac{1}{\rho}$.

$$\boxed{J = \sigma \vec{E}}$$

Calculating Resistance from Resistivity

(*) Resistance is a property of an object. Resistivity is a property of a material.

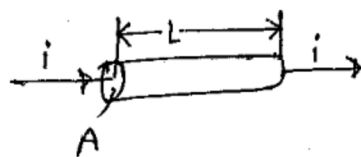
Q If we know the resistivity of a substance such as copper, we can calculate resistance of a length of wire made of that substance.

As $dE = \frac{dV}{ds}$, $E = \frac{V}{L}$ and $J = \frac{i}{A}$

$$\begin{aligned} \rho &= \frac{E}{J} = \frac{V/L}{i/A} \\ &= \frac{V}{L} \times \frac{A}{i} \\ &= \frac{V}{i} \left(\frac{A}{L} \right) \end{aligned}$$

$$\rho = R \frac{A}{L}$$

$$\boxed{R = \rho \frac{L}{A}}$$

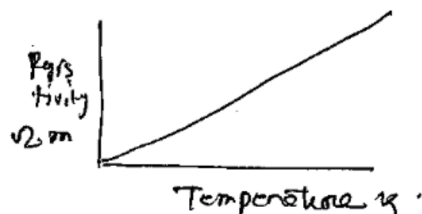


The macroscopic quantities V , i , and R are of greatest interest as we frequently measure them.

→ We turn to microscopic quantities E , J and ρ when we are interested in the fundamental electrical properties of materials.

Temperature effects on resistivity

The relation between temperature and resistivity for copper — and for metals in general — is fairly linear over a rather broad temperature range.

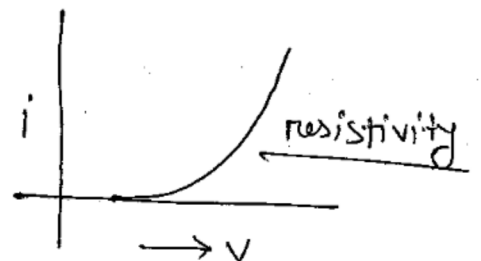
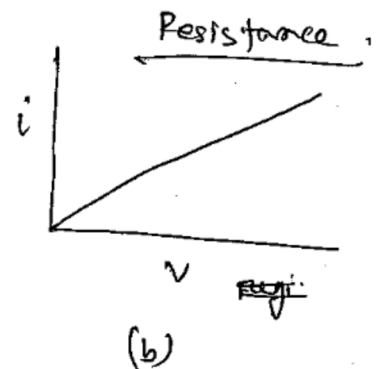
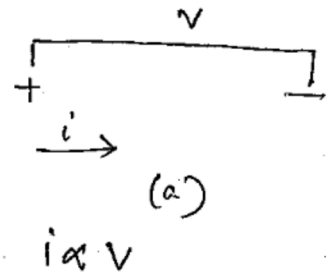


OHM'S LAW

- (a) ~~Q~~ Current through a device is always directly proportional to the potential difference applied to the device.
- (b) ~~Q~~ A conducting device obeys Ohm's law when the resistance of the device is independent of the magnitude and polarity of the applied potential difference.
- (c) ~~Q~~ A conducting material obeys Ohm's law when the resistivity of the material is independent of the magnitude and direction of the applied electric field.

$$V = IR$$

$$E = \rho \vec{J}$$



A Microscopic View of Ohm's Law

Assumption closer to the quantum reality is that conduction electrons in a metal move with a single effective speed v_{eff} , and this speed is essentially independent of the temperature. For Copper $v_{eff} \approx 1.6 \times 10^6$ m/s.

When we apply an electric field to a metal sample, the electrons modify their random motions slightly and drift very slowly — in a direction opposite that ~~the~~ of the field — with an average drift speed v_d .

→ The drift speed in a typical metallic conductor is about 5×10^{-7} m/s, less than the effective speed 1.6×10^6 m/s.

The motion of conduction electrons in an electric field $\vec{E}$ is thus a combination of the motion due to random collisions and that due to $\vec{E}$.

When we consider all the free electrons, their random motions average to zero and make no contribution to the drift speed. Thus, the drift speed is due only to the effect of the electric field on the electrons.

Classically If an electron of mass m is placed in an electric field of magnitude E , the electron will experience an acceleration given by Newton's second law.

$$a = \frac{F}{m} = \frac{eE}{m}$$

If τ_{time} between collisions, the average electron will acquire a drift speed of $v_d = a\tau /$

Moreover, if we measure the drift speeds of all the electrons at any instant, we will find that their average drift speed is also $a\tau$.

$$v_d = a\tau = \frac{eE\tau}{m}$$

As $\vec{J} = ne\vec{v}_d$, we get $v_d = \frac{\vec{J}}{ne} = \frac{eE\tau}{m}$

$$\therefore E = \left(\frac{m}{e^2 n \tau} \right) \vec{J}$$

again comparing with $E = \rho \vec{J}$

$$\boxed{\rho = \frac{m}{e^2 n \tau}}$$

$v_d \ll v_{rms}$ / τ is independent of the strength of the applied electric field.

we can say $\rho \approx \text{constant}$ and independent of applied electric field strength and metals obey Ohm's law

26.05 If mean free time $\tau = 2.5 \times 10^{-14} \text{ s}$ between collisions for the conduction electrons in copper. What is λ (mean free path) of the conduction electrons in a conductor. Assuming effective speed v_{eff} is $1.6 \times 10^6 \text{ m/s}$.

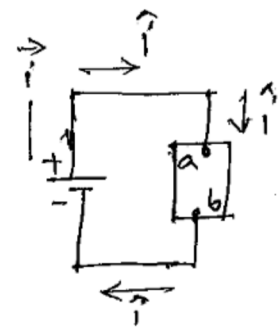
$$\lambda = v_{eff} \tau$$

$$= (1.6 \times 10^6 \text{ m/s}) (2.5 \times 10^{-14} \text{ s})$$

$$= 4.0 \times 10^{-8} \text{ m} = 40 \text{ nm}$$

This is about 150 times the distance between nearest neighbour atoms in a copper lattice. Each conduction electron passes many copper atoms, before finally ~~the~~ hitting one.

Power in Electric Circuits



$$dU = dq V = i dt V$$

The principle of conservation of energy tells us that the decrease in electrical potential energy from a to b is accompanied by a transfer of energy to some other form.

$$P \Rightarrow \frac{dU}{dt} = iV \quad (\text{rate of electrical energy transfer})$$

$$P = \text{Volt-ampere} = \left(1 \frac{V}{C}\right) \left(1 \frac{C}{s}\right) = 1W.$$

$$P = i^2 R \quad (\text{resistive dissipation})$$

$$= \frac{V^2}{R} \quad (\text{resistive dissipation})$$

Semiconductor

$$n = \frac{n}{e^2 \tau} \rightarrow \text{number of carriers}$$

In semiconductor, n is small but increases very rapidly with temperature as the increased thermal agitation makes more charge carriers available.

Superconductor

4K $\rightarrow$ without thermal energy.